



“Yin-Yang grid”: An overset grid in spherical geometry

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[1] A new kind of overset grid, named Yin-Yang grid, for spherical geometry is proposed. The Yin-Yang grid is composed of two identical component grids that are combined in a complementary way to cover a spherical surface with partial overlap on their boundaries. Each component grid is a low-latitude part of the latitude-longitude grid. Therefore the grid spacing is quasi-uniform, and the metric tensors are simple and analytically known. One can directly apply mathematical and numerical resources that have been written in the spherical polar coordinates or latitude-longitude grid. The complementary combination of the two identical component grids enables us to make efficient and concise programs. Simulation codes for geodynamo and mantle convection simulations using finite difference scheme based on the Yin-Yang grid are developed and tested. The Yin-Yang grid is suitable for massively parallel computers.

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1. Introduction

[2] Since the Earth is composed of spherical layers, computer simulations of the Earth’s interior, such as geodynamo and mantle convection simulations, need efficient spatial discretization schemes in spherical shell geometry. The spectral method [Glatzmaier, 1984] has been the major tool in the geodynamo simulation; all six codes [Dormy *et al.*, 1998; Christensen *et al.*, 1999; Sakuraba and Kono, 1999; Tilgner, 1999; Takahashi *et al.*, 2001] in the benchmark test of Christensen *et al.* [2001] and other codes [e.g., Kuang and Bloxham, 1999; Ishihara and Kida, 2002] use the spherical harmonics expansion method in the horizontal space. However, the importance of nonspectral (or point-based) approaches in the dynamo simulation is now increasingly recognized to simulate more realistic geodynamo regime with smaller Ekman numbers [Chan *et al.*, 2001]. The pursuit of point-

based approaches started earlier in the mantle convection simulations, because the mantle’s intense spatial variation of viscosity and the phase transitions makes the spectral approach not fit to the problem. Although the spectral method for the mantle convection prospered in 1980s and 1990s [Machetel *et al.*, 1986; Glatzmaier, 1988; Bercovici *et al.*, 1989; Zhang and Yuen, 1995; Harder and Christensen, 1996], the finite element method is rapidly growing in this field [Baumgardner, 1985; Bunge and Baumgardner, 1995; Zhong *et al.*, 2000; Tabata and Suzuki, 2000; Richards *et al.*, 2001]. There are also a couple of codes that uses the finite element method in the geodynamo simulation [Chan *et al.*, 2001; Matsui and Okuda, 2002]. The finite difference or finite volume method is applied for the mantle convection by Ratcliff *et al.* [1996], Iwase [1996], and Hernlund and Tackley [2003]. The finite difference method has been used for the core convection and the geodynamo simu-

lation in studies from 1990s [Kageyama *et al.*, 1993, 1995; Kageyama and Sato, 1997a, 1997b, 1997c; Kageyama *et al.*, 1999; Ochi *et al.*, 1999; Li *et al.*, 2002], in which the latitude-longitude grids in the spherical polar coordinates is used with radius r ($r_i \leq r \leq r_o$), colatitude θ ($0 \leq \theta \leq \pi$), and longitude ϕ ($0 \leq \phi < 2\pi$). Since the finite difference method enables us to make highly optimized programs for massively parallel computers, especially massively parallel vector supercomputers like the Earth simulator [Habata *et al.*, 2003], we further exploit the possibility of the finite difference method for simulations in spherical shell geometry by improving the base grid system.

[3] It is known that the latitude-longitude grid has two numerical problems; the coordinate singularity and the grid convergence near the poles. Since the coordinate singularity is not a real singularity (the pole is not singular point of physical functions), one can solve the basic equations on the poles by applying the l'Hospital's rule on the pole grids [e.g., Kageyama *et al.*, 1995]. The computational cost for this pole grid solver is negligible.

[4] The problem of the grid convergence is more serious. In order to relax the severe restriction on the time step, one has to apply a filter so that the grid spacing on the sphere becomes effectively quasi-uniform. The amount of information abandoned by the filter is estimated by the number of grid points that are effectively present and that actually present in the computational space; suppose one has a latitude-longitude grid of a spherical surface of unit radius with inter mesh angles Δ in both colatitude (θ) and longitude (ϕ). The azimuthal grid spacing, which is Δ in the equator, converges in higher latitudes. When a filter enables an effectively quasi-uniform grid with spacing Δ on the sphere, the number of effective grid points is estimated by $(\int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi)/\Delta^2 = 4\pi/\Delta^2$. While the number of actual grid points in the computational space is given by $(\int_0^\pi d\theta \times \int_0^{2\pi} d\phi)/\Delta^2 = 2\pi^2/\Delta^2$. Therefore sizable ratio of information, $(2\pi^2 - 4\pi)/2\pi^2 \sim 36\%$ of the latitude-longitude grid, is abandoned in vain by the filtering at each simulation step. In addition to this computational inefficiency, the filter has nonnegligible computational costs. In our geodynamo simulation code using latitude-longitude grid, in which a fast Fourier transform (FFT)-based filtering procedure is applied, the filter routine can take more than 30% of the total execution time.

[5] Note that the above problem of the grid redundancy in the latitude-longitude grid comes only

from the region of high latitudes. The remaining part of the latitude-longitude grid, the low-latitude region, has rather desirable feature for numerical simulations; it is an orthogonal grid, it has simple metric tensors, and it has quasi-uniform grid spacings. This observation leads us to the idea of a new spherical grid proposed in this paper.

[6] Since there is no grid mesh that is orthogonal all over the spherical surface and, at the same time, free of coordinate singularity or grid convergence, we decompose the spherical surface into subregions. The decomposition, or dissection, enables us to cover each subregion by a grid system that is individually orthogonal and singularity-free. This divide-and-rule approach has been used with good success in the computational aerodynamics that incorporates complex geometry of aircraft's body with wings/stores/blades.

[7] The dissection of the computational domain generates internal border or boundary between the subregions. There are two different approaches to handle the internal boundaries. One is the patched grid method [Rai, 1986] and the other is the overset grid method [Chesshire and Henshaw, 1990]. In the patched grid approach, the subdomains contact one another without any overlap on their borders. In the overset grid method, on the other hand, the subdomains partially overlap one another on their borders. The overset grid is also called as overlaid grid, or composite overlapping grid, or Chimera grid [Steger *et al.*, 1983]. The validity and importance of the overset approach in the aerodynamical calculations was pointed out by Steger [1982]. Since then this method is widely used in this field. It is now one of the most important grid techniques in the computational aerodynamics; for example, whole aircraft with wing and store [Meakin, 1992], tiltrotor aircraft [Meakin, 1993], Boeing 747 [Cao *et al.*, 1998; Rogers *et al.*, 1998], space shuttle [Buning *et al.*, 1988], helicopter [Duque *et al.*, 1996], and others.

[8] In the computational geosciences, the idea of the overset grid approach appeared rather early. N. Phillips proposed a kind of composite grid in 1950s to solve partial differential equations on a hemisphere, in which the high-latitude region of the latitude-longitude grid is "capped" by another grid system that is constructed by a stereographic projection to a plane on the north pole [Phillips, 1957, 1959; Browning *et al.*, 1989]. After a long intermission, the overset grid method seems to

attract growing interest in geoscience these days. The “cubed sphere” [Ronchi *et al.*, 1996] is an overset grid that covers a spherical surface with six component grids that correspond to six faces of a sphere. The “cubed sphere” is recently applied to the mantle convection simulation [Hernlund and Tackley, 2003]. In the atmospheric research, other kind of spherical overset grid is used in a global circulation model [Dudhia and Bresch, 2002], in which the spherical surface is covered by two-component grids, improved stereographic projection grids, in northern and southern hemispheres that overlap in the equator. A successful test of 100-day integration of global circulation is demonstrated with this overset grid.

[9] The overset grid proposed in this paper is named “Yin-Yang grid” after the symbol for yin and yang of Chinese philosophy of complementarity. The Yin-Yang grid is composed of two identical and complementary component grids. Compared with other spherical overset grids, the Yin-Yang grid is simple in its geometry and metric tensors. A remarkable feature of this overset grid is that the two identical component grids are combined in a complementary way with a special symmetry.

2. Basic Yin-Yang Grid

[10] The Yin-Yang grid in its most basic shape is shown in Figure 1. It has two component grids that are geometrically identical (exactly the same shape and size); see Figure 1a. We call the two component grids “Yin grid” (or n grid) and “Yang grid” (or e grid). They are combined to cover a spherical surface with partial overlap on their borders as shown in Figure 1b. Each component grid is in fact a part of the latitude-longitude grid: A component grid, say Yin grid, is defined in the spherical polar coordinates by

$$(\pi/4 - \delta \leq \theta \leq 3\pi/4 + \delta) \cap (-3\pi/4 - \delta \leq \phi \leq 3\pi/4 + \delta), \quad (1)$$

where δ is a small buffer, which is proportional to grid spacing, required for minimum overlap in the overset methodology (see Figure 1b). In the limit of infinitesimal grid ($\delta \rightarrow 0$), the area of the above part of the sphere with unit radius is given by $\int_{\pi/4}^{3\pi/4} \sin \theta d\theta \int_{-3\pi/4}^{3\pi/4} d\phi = 3\pi/\sqrt{2} \sim 2.12\pi$, i.e., roughly a half of the whole spherical surface (2π). Another component grid, Yang grid, is defined by the same rule of equation (1) but in different spherical coordinates that is perpendicular to the

original one; see the green- and blue-colored spherical mesh in Figure 2a. The axis of the Yang grid’s coordinates (blue mesh in Figure 2a) is located in an equator of the Yin grid’s coordinates (green mesh in Figure 2a). The relation between Yin coordinates and Yang coordinates is denoted in the Cartesian coordinates by

$$(x^e, y^e, z^e) = (-x^n, z^n, y^n), \quad (2)$$

where (x^n, y^n, z^n) is Yin’s Cartesian coordinates and (x^e, y^e, z^e) is Yang’s. In a matrix form,

$$\begin{pmatrix} x^e \\ y^e \\ z^e \end{pmatrix} = M \begin{pmatrix} x^n \\ y^n \\ z^n \end{pmatrix}, \quad (3)$$

where

$$M = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (4)$$

Note that

$$M^{-1} = M, \quad (5)$$

which indicates that the transformations between Yin and Yang coordinates are symmetric. This is a reflex of the complementary relation between Yin and Yang.

[11] In the spherical coordinates, equation (2) reads

$$r^e = r^n, \quad (6)$$

$$\sin \theta^e \cos \phi^e = -\sin \theta^n \cos \phi^n, \quad (7)$$

$$\sin \theta^e \sin \phi^e = \cos \theta^n, \quad (8)$$

$$\cos \theta^e = \sin \theta^n \sin \phi^n, \quad (9)$$

where (r^n, θ^n, ϕ^n) , and (r^e, θ^e, ϕ^e) are the coordinates of Yin and Yang, respectively. The idea of two perpendicular spherical coordinates is used in the global ocean simulation [Eby and Holloway, 1994] to avoid the grid convergence in the Arctic, however, the second spherical coordinates is used in a sort of auxiliary way for the main (usual) spherical polar coordinates in their method. On the other hand, we make the best use of the symmetry between two coordinates.

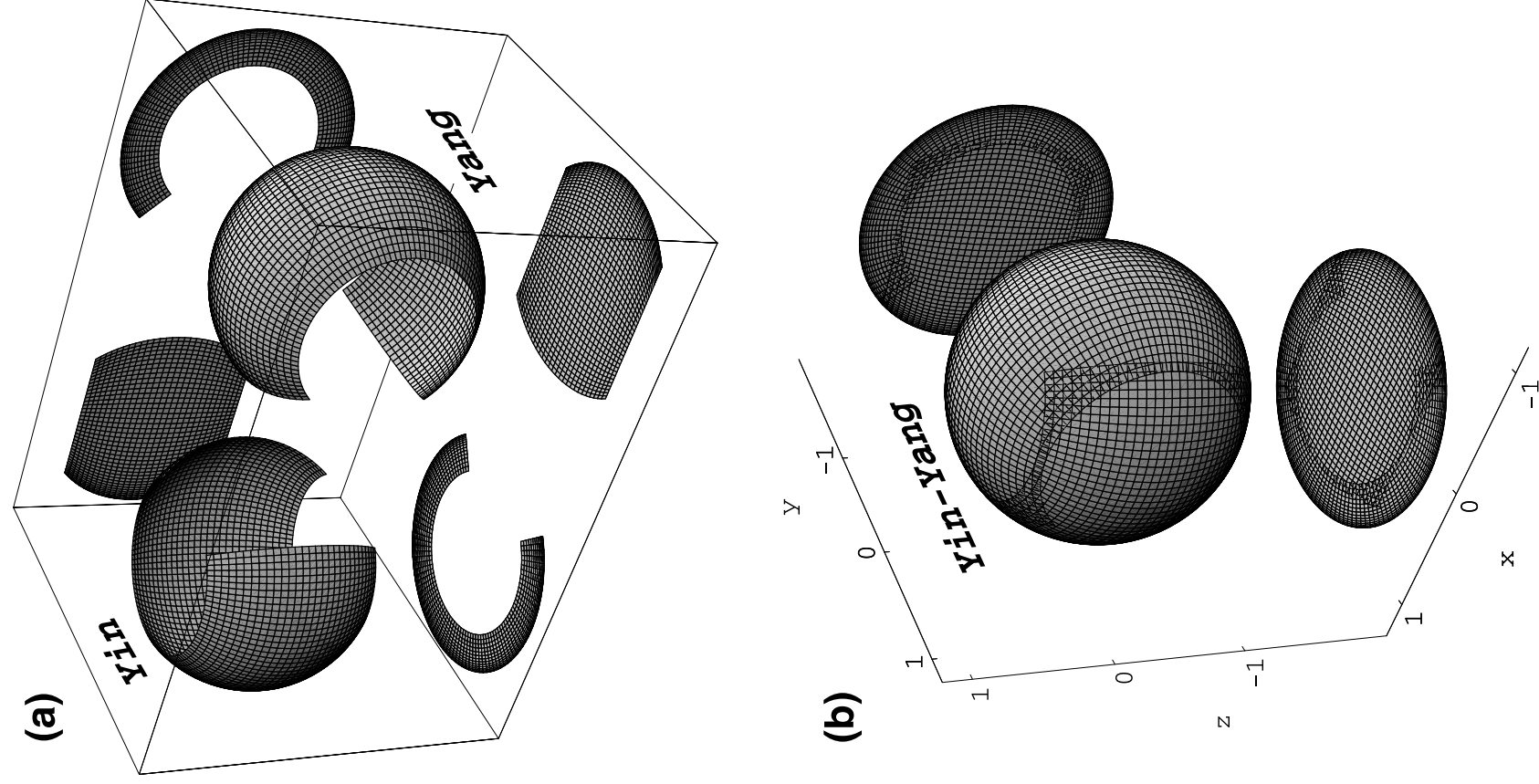


Figure 1. Basic Yin-Yang grid. (a) Two identical component grids, named Yin and Yang. (b) Yin grid and Yang grid combined to cover a spherical surface with partial overlap.

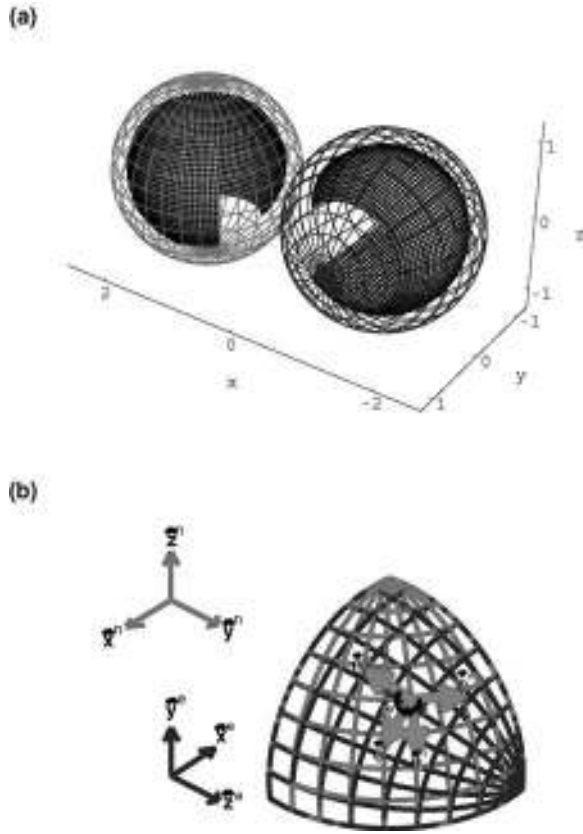


Figure 2. (a) Component grid of the Yin-Yang grid as a part of the latitude-longitude grid. The axes of two spherical polar coordinates for Yin grid (green mesh) and Yang grid (blue mesh) are perpendicular. (b) Unit horizontal vectors of Yin and Yang coordinates. They are mapped by a local rotation transform with angle ψ .

[12] For spatial discretization, we define mesh point at j th colatitude θ_j^ℓ and k th longitude ϕ_k^ℓ on Yin grid (for $\ell = n$) and on Yang grid (for $\ell = e$) as

$$\theta_j^\ell = \theta_{\min} + j \Delta\theta, \quad (j = 0, N_\theta - 1), \quad (10)$$

$$\phi_k^\ell = \phi_{\min} + k \Delta\phi, \quad (k = 0, N_\phi - 1), \quad (11)$$

with

$$\Delta\theta = (\theta_{\max} - \theta_{\min}) / (N_\theta - 1), \quad (12)$$

$$\Delta\phi = (\phi_{\max} - \phi_{\min}) / (N_\phi - 1), \quad (13)$$

where the grid distribution ranges from $\theta_{\min} = \pi/4 - \delta$ to $\theta_{\max} = 3\pi/4 + \delta$ in colatitude, and from $\phi_{\min} = -3\pi/4 - \delta$ to $\phi_{\max} = 3\pi/4 + \delta$ in longitude. We set $\Delta\theta = \Delta\phi = 2\delta$ in Figure 1, as an example.

[13] An important feature of the Yin-Yang grid as a spherical overset grid is that the two component grids are identical and their geometrical positions

are complementary. This enables us to make concise programs: Suppose a grid point (θ_j^n, ϕ_k^n) on Yin grid's horizontal border at index position (j, k) (e.g., $j = 1$). Its value should be determined by an interpolation from its neighbor points, or stencils, of Yang grid with interpolation coefficients that are determined by relative position of (θ_j^n, ϕ_k^n) in the stencils. Note that exactly the same interpolation coefficients and relative stencils are used to set the value of corresponding grid point (θ_j^e, ϕ_k^e) at (j, k) of Yang's border, since the geometrical relations between Yin grid and Yang grid are symmetric. In other words, we can make use of one interpolation routine for two times (for Yin grid and for Yang grid) to set the horizontal boundary conditions. Note also that the metric tensors at a bulk grid point at (j, k) of Yin grid is a function of its position (θ_j^n, ϕ_k^n) in Yin's coordinates, and the metric tensors at corresponding point (θ_j^e, ϕ_k^e) in Yang grid are exactly the same. Therefore we can call one subroutine of fluid solver and others for two times for Yin grid and Yang grid.

[14] Another advantage of the Yin-Yang grid resides in the fact that the component grid is nothing but the (part of) latitude-longitude grid. We can directly deal with the equations to be solved with the vector form in the usual spherical polar coordinates, $\{v_r, v_\theta, v_\phi\}$. The analytical form of metric tensors are familiar in the spherical coordinates. We can directly code the basic equations in the program as they are formulated in the spherical coordinates. We can make use of various resources of mathematical formulas, program libraries, and tools that have been developed in the spherical polar coordinates.

[15] To conclude this section, we point out that the construction of three-dimensional Yin-Yang grid for spherical shell geometry is straightforward, by piling up the basic (two-dimensional) Yin-Yang grids in radial direction (see Figure 3).

3. Vector Transformation Formula Between Yin and Yang Grids

[16] Following the general overset methodology [e.g., Chesshire and Henshaw, 1990], interpolations are applied on the boundary of each component grid to set the boundary values, or internal boundary condition. When one deals with scalar variables, the interpolation is simple. For vector fields, a care is needed for vector components, since expressions of a vector in the Yin's spherical

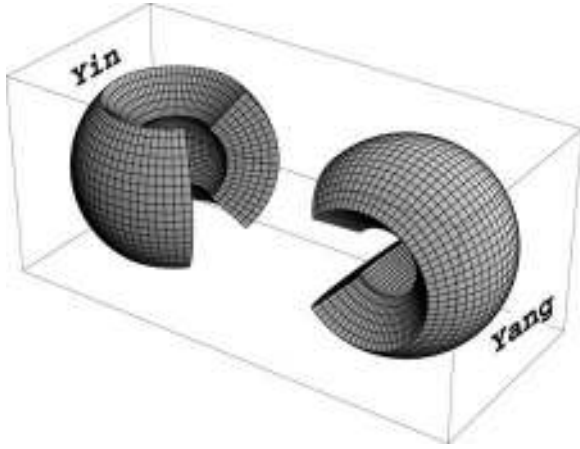


Figure 3. Three-dimensional Yin-Yang grid for spherical shell geometry. This is constructed by piling up the basic Yin-Yang grid shown in Figure 1 in the radial direction.

coordinates, $\{v_r^n, v_\theta^n, v_\phi^n\}$, and in the Yang's coordinates, $\{v_r^e, v_\theta^e, v_\phi^e\}$, are different.

[17] Because the Yin-Yang transformation denoted by equation (2) is a rotation about the origin ($r = 0$), the radial component of the vector is invariant ($v_r^n = v_r^e$), and horizontal components are mapped by local rotation transforms, as shown in Figure 2b, where the rotation angle ψ is a function of latitude and longitude;

$$\begin{pmatrix} v_r^e \\ v_\theta^e \\ v_\phi^e \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} v_r^n \\ v_\theta^n \\ v_\phi^n \end{pmatrix}. \quad (14)$$

To find the expression of ψ , we consider unit vectors in θ and ϕ directions on the Yin and Yang coordinates. From Figure 2b, we see

$$\cos \psi = \hat{\phi}^n \cdot \hat{\phi}^e, \quad (15)$$

$$\sin \psi = -\hat{\phi}^n \cdot \hat{\theta}^e, \quad (16)$$

where $\hat{\theta}^e$ and $\hat{\phi}^e$ are unit vectors in θ and ϕ directions in the component grid ℓ , with $\ell = n$ for Yin grid, and $\ell = e$ for Yang grid. The unit vectors $\hat{x}^\ell, \hat{y}^\ell, \hat{z}^\ell$ in the Cartesian coordinates are related to $\hat{\theta}^\ell, \hat{\phi}^\ell$ by

$$\hat{\phi}^e = -\sin \phi^e \hat{x}^e + \cos \phi^e \hat{y}^e, \quad (17)$$

$$\begin{aligned} \hat{\phi}^n &= -\sin \phi^n \hat{x}^n + \cos \phi^n \hat{y}^n \\ &= \sin \phi^n \hat{x}^e + \cos \phi^n \hat{z}^e, \end{aligned} \quad (18)$$

$$\hat{\theta}^e = \cos \theta^e \cos \phi^e \hat{x}^e + \cos \theta^e \sin \phi^e \hat{y}^e - \sin \theta^e \hat{z}^e. \quad (19)$$

Substituting equations (17) and (18) into (15), we get

$$\cos \psi = -\sin \phi^e \sin \phi^n. \quad (20)$$

Substituting equations (18) and (19) into (16), we get

$$\begin{aligned} \sin \psi &= \cos \theta^e \cos \phi^e \sin \phi^n - \sin \theta^e \cos \phi^n \\ &= \frac{1}{\sin \theta^e \sin \theta^n} \{ \cos \theta^e (\sin \theta^e \cos \phi^e) (\sin \theta^n \sin \phi^n) \\ &\quad - \sin^2 \theta^e (\sin \theta^n \cos \phi^n) \} \\ &= -\frac{\cos \phi^e}{\sin \theta^n} \\ &= \frac{\cos \phi^n}{\sin \theta^e}. \end{aligned} \quad (21)$$

Here we have used equations (7)–(9).

[18] From equations (20), (21), (15), (16) and (14), we obtain the transformation formula of the vector components (v_r, v_θ, v_ϕ) by

$$\begin{pmatrix} v_r^e \\ v_\theta^e \\ v_\phi^e \end{pmatrix} = P \begin{pmatrix} v_r^n \\ v_\theta^n \\ v_\phi^n \end{pmatrix}, \quad (22)$$

with the transformation matrix

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\sin \phi^e \sin \phi^n & -\cos \phi^n / \sin \theta^e \\ 0 & \cos \phi^n / \sin \theta^e & -\sin \phi^e \sin \phi^n \end{pmatrix}. \quad (23)$$

Since Yin and Yang coordinates are symmetric, the inverse transformation from Yang into Yin is given by the interchange of the suffixes:

$$P^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\sin \phi^n \sin \phi^e & -\cos \phi^e / \sin \theta^n \\ 0 & \cos \phi^e / \sin \theta^n & -\sin \phi^n \sin \phi^e \end{pmatrix}. \quad (24)$$

Note also that

$$P^2 = 1, \quad (25)$$

which indicates the complementary relation between Yin and Yang coordinates.

[19] When we see the component grid of the basic Yin-Yang grid shown in Figure 1 in the Mercator projection, it is a rectangle; the four corners intrude most into the other component

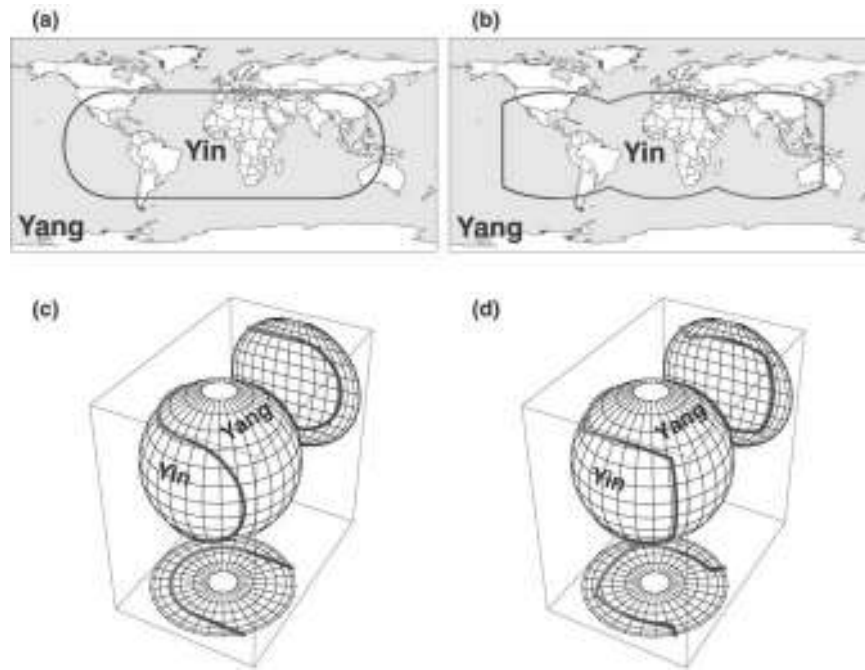


Figure 4. (a and b) Curves that divide a spherical surface into two identical areas. If one cuts along the blue-red curve, the spherical surface is divided into two parts (denoted by Yin and Yang in the pictures) that are exactly the same shape and size. The blue part of the curve and red part of the curve are in the complementary relation; the blue curve of Yin is red curve of Yang, and vice versa. (c and d) Corresponding three-dimensional views.

grid (see Figure 1b). Even if the grid mesh is taken to be infinitesimal, i.e., $\Delta\theta = \Delta\phi \rightarrow 0$ and $\delta \rightarrow 0$, the overlapping area has still nonzero ratio of about 6%; $(3/\sqrt{2}\pi - 2\pi)/2\pi \sim 0.061$. Following the general overset grid methodology, we take the interpolation on the boundary of each grid, i.e., on the border of the rectangles, and the interpolated data are used as the boundary condition. We obtain two solutions, one from Yin side and other from Yang side, in the overlapped area. This might be a slight (6%) waste of computational time, but the “double solution” causes no problem in actual calculations. The difference between the two solutions is within the discretization error that is omnipresent all over the sphere any way.

[20] The overlapped area can be minimized by modifying the component grid's shape from the rectangle. It is obvious that a Yin-Yang grid with minimum overlap region can be constructed by a division, or dissection, with a closed curve on a sphere that cuts the sphere into two identical parts. There are infinite number of such dissections of a sphere. Figures 4a and 4b show two examples. When we cut along the curve that is colored with red and blue in Figure 4a or

Figure 4b, we get two separated parts of the spherical surface that are identical. Although, it is not apparent that the two parts separated by the blue-red curve in Figures 4a and 4b are identical, the corresponding three-dimensional view (Figures 4c and 4d) shows this more convincingly. The cutoff curve of Figure 4c reminds us of a baseball, while the cutoff curve of Figure 4d resembles a cube.

[21] On the basis of these spherical dissections we can construct spherical overset grids with two identical component grids that has minimum overlapping area; Figure 5 shows a Yin-Yang grid that corresponds to the baseball-type dissection of a sphere (Figures 4a and 4c). Figure 6 is for the cube-type dissection (Figures 4b and 4d). When minimizing the computational cost is strongly required, the Yin-Yang grid of the baseball type (Figure 5) or cube type (Figure 6) would be worth trying.

[22] However, the nonrectangle geometries of the component grid of Figures 5 and 6 imply that special cares should be taken to mask some grid points. The number of the mask is the same for both the Yin-Yang grids of Figures 5 and 6 since the nonmasked area of a component grid is just a

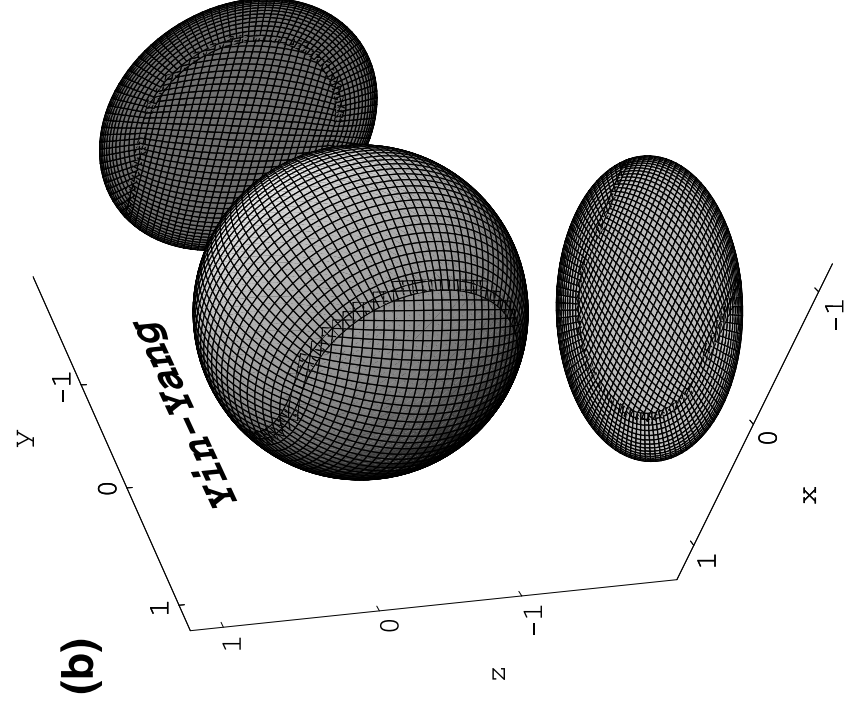
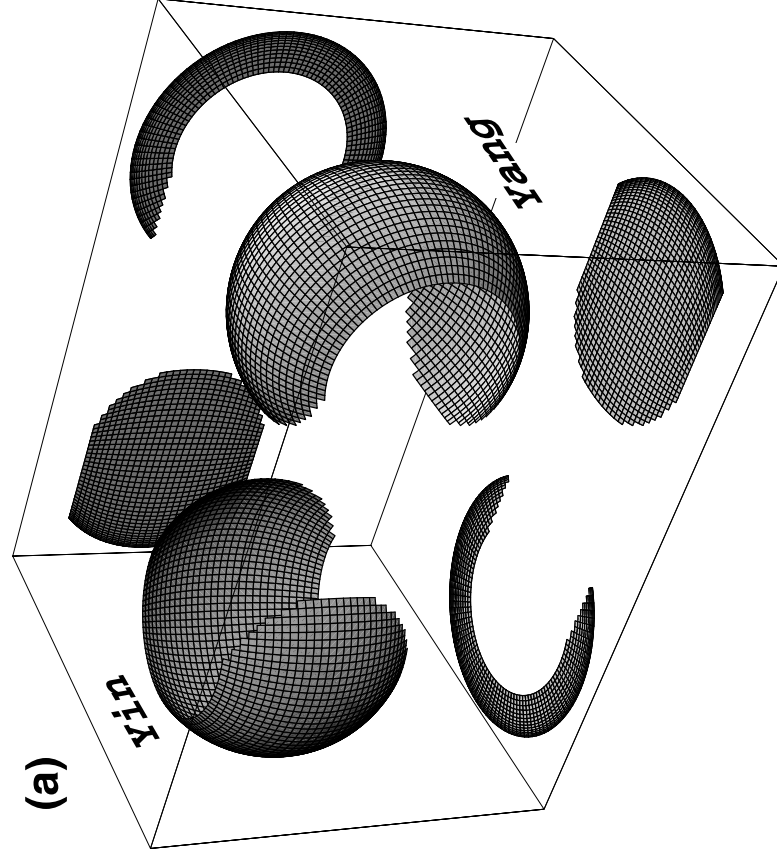


Figure 5. (a and b) A Yin-Yang grid with minimum overlap that has the baseball-like border curve between Yin and Yang grids. Corresponding spherical dissection is given in Figures 4a or 4c.

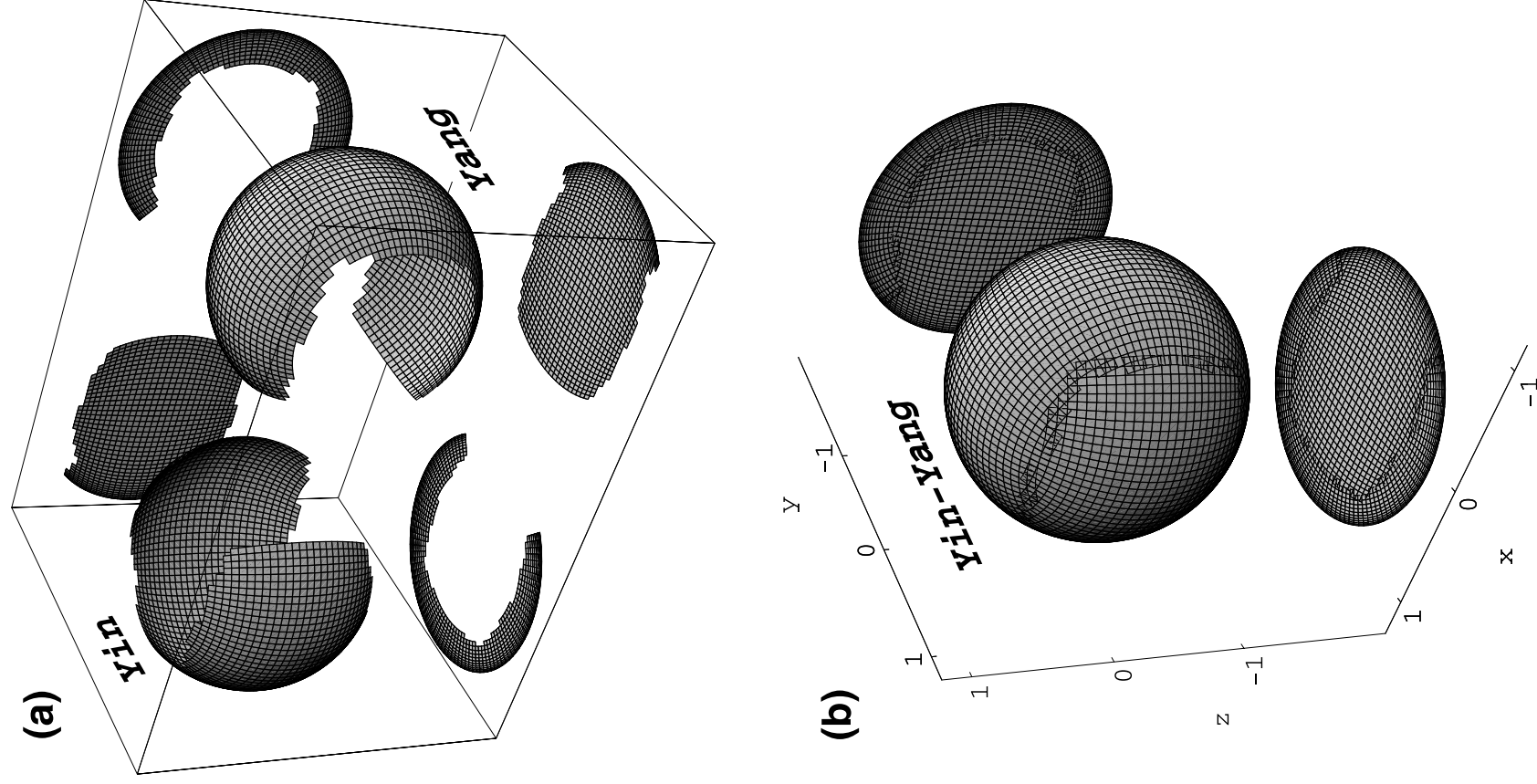


Figure 6. (a and b) Another Yin-Yang grid with minimum overlap. The border curve between Yin and Yang grids is cube-like. Corresponding spherical dissection is given in Figures 4b or 4d.

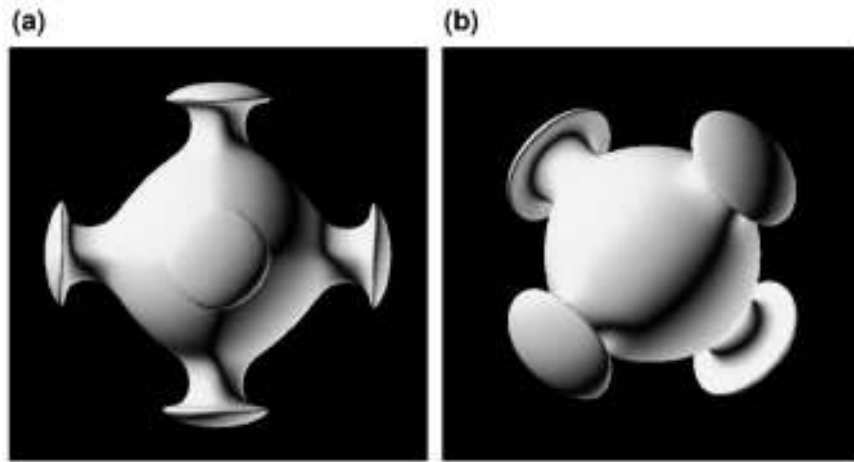


Figure 7. Isosurfaces of temperature of mantle convection simulation by Yin-Yang grid at Rayleigh number $R = 10^4$. Stationary solutions with (a) cubic symmetry and (b) tetrahedral symmetry.

half of the spherical surface (2π) in the limit of the negligibly small overlap area.

4. Applications and Validation Tests of Yin-Yang Grid

[23] To confirm the validity and the scalability on parallel computers, we applied the Yin-Yang grid to two kinds of the solid earth simulation; mantle convection simulation and geodynamo simulation. In both cases, the basic Yin-Yang grid described in section 2 was used for the spatial discretization in spherical shell geometry. The calculations were performed on the Earth Simulator [Shingu *et al.*, 2002; Yokokawa *et al.*, 2002; Habata *et al.*, 2003], which is a massively parallel supercomputer of distributed memory type. It consists of 5120 vector-type processors. The peak performance of each processor is 8 Gflops; the Earth simulator as a whole has 40 Tflops of total peak performance.

4.1. Mantle Convection Simulation

[24] We have applied the basic Yin-Yang grid to a mantle convection simulation in which we solve time development of thermal convection motion of a Boussinesq fluid with infinite Prandtl number in a spherical shell. The basic equations are

$$\nabla \cdot \mathbf{v} = 0, \quad (26)$$

$$0 = -\nabla p + \nabla \cdot (\mu \epsilon) + R T \hat{\mathbf{r}}, \quad (27)$$

$$\frac{\partial T}{\partial t} = \nabla^2 T - \mathbf{v} \cdot \nabla T, \quad (28)$$

where $\mathbf{v}$ is the velocity, p pressure, T temperature, ϵ strain rate tensor, R Rayleigh number, and $\hat{\mathbf{r}}$ is unit vector in radial direction. We assume that viscosity μ depends only on temperature; $\mu(T) \propto \exp[-h(T - 0.5)]$, where h denotes the degree of viscosity contrast between the top and bottom spherical boundaries on which temperatures are uniform and normalized as $T = 0$ and 1, respectively.

[25] We used second-order finite difference method on the Yin-Yang grid with the total grid size of 102 (in the radial direction r) $\times$ 54 (in the colatitudinal direction θ with $\pi/4 \leq \theta \leq 3\pi/4$) $\times$ 158 (in the longitudinal direction ϕ with $-3\pi/4 \leq \phi \leq 3\pi/4$) $\times$ 2 (for Yin and Yang). We used the Crank-Nicholson method for the time integration of equation (28) and the SIMPLER method to obtain $\mathbf{v}$ and p from equations (26) and (27).

[26] We first confirmed that our code reproduces the stationary and symmetric solutions with the tetrahedral and cubic symmetries at low Rayleigh number R . These solutions were found by linear theory [Busse, 1975; Busse and Riahi, 1982] and confirmed by numerical simulations [Bercovici *et al.*, 1989; Ratcliff *et al.*, 1996]. The isosurfaces of temperature at $R = 10^4$ are shown in Figure 7a for the cubic symmetry and in Figure 7b for the tetrahedral symmetry.

[27] Then, we performed benchmark tests with previously reported eight numerical mantle convection simulations [Glatzmaier, 1988; Bercovici *et al.*, 1989; Harder and Christensen, 1996; Ratcliff *et al.*, 1996; Iwase, 1996; Zhong *et al.*, 2000; Tabata and Suzuki, 2000; Richards *et al.*, 2001], in which various numerical schemes are adopted; the spectral method [Glatzmaier, 1988; Bercovici *et*

al., 1989; Harder and Christensen, 1996], the finite volume method [Ratcliff *et al.*, 1996; Iwase, 1996], and the finite element method [Tabata and Suzuki, 2000].

[28] We performed simulations with uniform ($h = 0$) and variable ($\log h = 20$) viscosity cases for the tetrahedral and cubic steady symmetric convections. Changing the Rayleigh number R as 2×10^3 , 7×10^3 , 1.4×10^3 , 3.5×10^3 , 7×10^3 , and 1.4×10^4 , we monitored Nusselt number at the core-mantle boundary and the root-mean-square velocity of entire spherical shell. In spite of the differences of the discretization methods and numerical schemes, all the values obtained by our convection code based on the Yin-Yang grid agree well with the previously reported values within a few percent or even better. This confirms the validity of the Yin-Yang grid as a solver of the incompressible fluid equations (26)–(28). Details of the Yin-Yang mantle convection code and benchmark tests, as well as some preliminary results, are reported in other paper [Yoshida and Kageyama, 2004].

4.2. Geodynamo Simulation

[29] We also applied the Yin-Yang grid method to a geodynamo simulations in which time development of thermal convection with dynamo action of a magnetohydrodynamic (MHD) fluid in a rotating spherical shell (outer core) is solved. The physical model for our geodynamo simulation is described in our papers [e.g., Kageyama *et al.*, 1995]. An electrically conducting fluid is confined the spherical shell that rotates with an angular velocity Ω . We use the rotating frame of reference with this angular velocity. As in the case of the mantle convection simulation, the temperatures of both the inner and outer spheres are fixed; hot (inner) and cold (outer). The convection motion sets in when a random temperature perturbation is imposed at the beginning of the calculation and, at the same time, an infinitesimally small, random “seed” of magnetic field is given.

[30] The fundamental equations are the following normalized MHD equations:

$$\frac{\partial p}{\partial t} = -\nabla \cdot \mathbf{f}, \quad (29)$$

$$\begin{aligned} \frac{\partial \mathbf{f}}{\partial t} = & -\nabla \cdot (\mathbf{v}\mathbf{f}) - \nabla p + \mathbf{j} \times \mathbf{B} + \rho \mathbf{g} + 2\rho \mathbf{v} \times \Omega \\ & + \mu \left(\nabla^2 \mathbf{v} + \frac{1}{3} \nabla (\nabla \cdot \mathbf{v}) \right), \end{aligned} \quad (30)$$

$$\begin{aligned} \frac{\partial p}{\partial t} = & -\mathbf{v} \cdot \nabla p - \gamma p \nabla \cdot \mathbf{v} + (\gamma - 1) K \nabla^2 T + (\gamma - 1) \eta \mathbf{j}^2 \\ & + (\gamma - 1) \Phi, \end{aligned} \quad (31)$$

$$\frac{\partial \mathbf{A}}{\partial t} = -\mathbf{E}, \quad (32)$$

with $\mathbf{B} = \nabla \times \mathbf{A}$, $\mathbf{j} = \nabla \times \mathbf{B}$, $\mathbf{E} = -\mathbf{v} \times \mathbf{B} + \eta \mathbf{j}$, $p = \rho T$, $\mathbf{g} = -g_0/r^2 \hat{\mathbf{r}}$, $\Phi = 2\mu(\epsilon \cdot \epsilon - (\nabla \cdot \mathbf{v})^2/3)$. Here the mass density ρ , pressure p , mass flux density $\mathbf{f}$, magnetic field's vector potential $\mathbf{A}$ are the basic variables in the simulation. Other quantities, magnetic field $\mathbf{B}$, electric current density $\mathbf{j}$, and electric field $\mathbf{E}$ are treated as subsidiary fields. The ratio of the specific heats γ , viscosity μ , thermal conductivity K and electrical resistivity η are assumed to be constant. The vector $\mathbf{g}$ is the gravity acceleration with constant g_0 .

[31] The spatial derivatives in the above equations are discretized by second-order central finite difference on the basic Yin-Yang grid described in section 2. The fourth-order Runge-Kutta method is used for the temporal integration.

[32] We set all the three dissipation constants (viscosity μ , thermal conductivity K , and electrical resistivity η) 10 times smaller than the values in our previous simulation in which repeated magnetic dipole reversals were observed [Li *et al.*, 2002]. The Rayleigh number of the present calculation is 3.36×10^6 , and the Taylor number is 5.88×10^8 . Figure 8 shows snapshot visualizations of columnar convection structure of this MHD simulation under these parameters. The total grid size for calculation of Figure 8 is 255 (in r) $\times$ 514 (in θ) $\times$ 1538 (in ϕ) $\times$ 2 (Yin and Yang). The dynamo-generated magnetic energy is saturated at the same level of the convection kinetic energy. Since details of the data analysis of this geodynamo simulation is beyond the scope of this paper, we focus here on aspects of the performance on parallel processing and its scalability.

[33] Because the Yin grid and Yang grid are identical, dividing the whole computational domain into Yin part and Yang part is not only natural but also efficient for parallel processing. In addition to this Yin and Yang division, further domain decompositions within each grid should be applied for massively parallel computers. There are several choices for the way of this domain decomposition. Here we take two-dimensional decomposition in the horizontal space, colatitude θ and longitude ϕ , in each Yin or Yang grid.

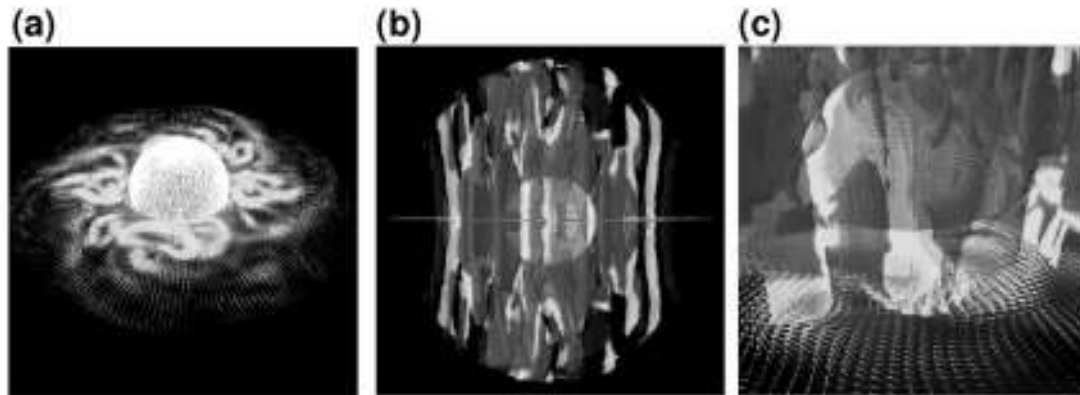


Figure 8. Snapshots of geodynamo simulation by Yin-Yang grid. The total grid size is 255 (radial) $\times$ 514 (latitudinal) $\times$ 1538(longitudinal) $\times$ 2(Yin and Yang). In these visualizations, grid points are reduced by a factor of 1/100. (a) Convection flow in the equatorial plane viewed from 45°N. (b) Columnar convection cells viewed in the equatorial plane. Two colors indicate cyclonic and anticyclonic convection columns. (c) Same data as Figure 8b, with closer view.

[34] In our geodynamo simulation code, we take the vector processing of the Earth Simulator in the radial dimension of three-dimensional (3-D) arrays of physical variables. We use MPI (Message-Passing Interface) library for parallel processing; one MPI process per one processor.

[35] When we used 1200 processors (1200 MPI processes) of the Earth Simulator with the total grid number of 255(r) $\times$ 514(θ) $\times$ 1538(ϕ) $\times$ 2(Yin and Yang), we achieved the performance of 5.40 Tflops. This is 56% of the theoretical peak performance. When we used 4096 processors (4096 MPI processes), we achieved 15.2 Tflops (46% of the peak performance) with the grid size of 511 (r) $\times$ 514(θ) $\times$ 1538(ϕ) $\times$ 2(Yin and Yang). We believe that this speed is the best record of geodynamo simulations reported so far. The above performance

and the scaling would demonstrate that the Yin-Yang grid is very suitable to parallel computers.

5. Summary

[36] For numerical simulations of the Earth's interior, we have developed a new spherical grid based on the overset grid methodology. Our motivation is to devise an spherical grid system that is suitable for finite difference scheme on massively parallel vector supercomputers. The spherical overset grid proposed in this paper, named Yin-Yang grid, is composed of two component grids. They have the same shape and size and combined to cover a spherical surface with partial overlap on their borders. Each component grid is nothing but low-latitude region of the usual latitude-longitude grid; it is 90° about the equator and 270° in the longi-

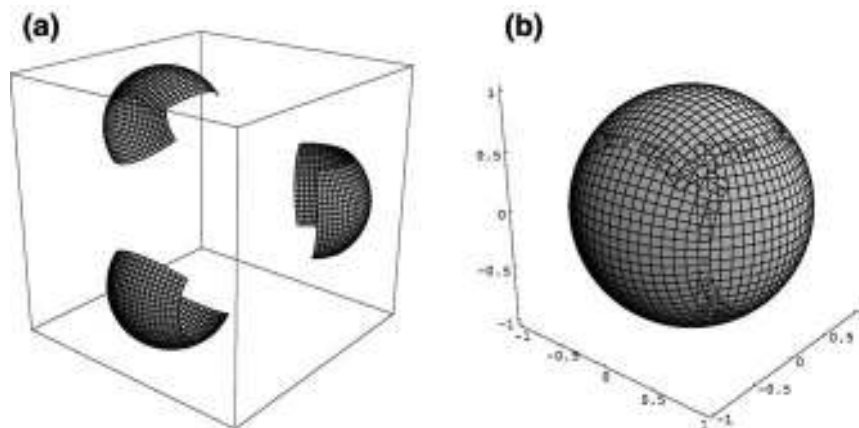


Figure 9. Another possible spherical overset grid that is composed of three identical component grids.

tude. Therefore the grid spacing is quasi-uniform and the metric tensors are simple and analytically known. One can directly apply mathematical and numerical resources that have been written in the spherical polar coordinates or latitude-longitude grid system. Since the two component grids are identical and combined in a complementary way, various routines for solvers and interpolation can be recycled for two times for each component grid at every simulation time step.

[37] We have developed finite difference codes of the mantle convection and dynamo simulation using the basic Yin-Yang grid for spherical shell geometry (see Figures 1 and 3). We have confirmed that the Yin-Yang grid is successfully applied to both cases. The mantle convection code is newly developed from scratch. Details of the code and simulation results are reported in other paper [Yoshida and Kageyama, 2004]; we solved the time development of thermal convection motion in a spherical shell of a Boussinesq fluid with infinite Prandtl number for uniform and variable viscosity cases. Our Yin-Yang mantle convection code successfully reproduced previously published results. The numerical values of Nusselt number and the mean velocity coincides with previously reported values within a few percent [Yoshida and Kageyama, 2004]. We have also applied the Yin-Yang grid to the geodynamo simulation code. The magnetohydrodynamic (MHD) equations with finite viscosity, thermal diffusivity, and electrical conductivity are solved. The Yin-Yang geodynamo code has been converted from our previous geodynamo code which was based on the latitude-longitude grid. We found that the code conversion was straightforward and rather easy since the base grid is common. We could reproduce our previous (latitude-longitude grid based) results of geodynamo simulation by our newly developed Yin-Yang geodynamo code with shorter calculation time. The details of the code will be reported in other paper.

[38] The Yin-Yang grid is suitable for parallel programming. Since the number of the component grid is two, we are naturally lead to make parallel programs with domain decomposition of even number: We first decompose whole computational region into two (Yin component and Yang component), then apply further domain decomposition in each component.

[39] Finally, we point out another possible spherical overset grid that has an odd number of component grids. Figure 9 shows a spherical overset

grid that consists of three identical component grids. In this case, the component grid is defined as a part (about 1/3) of the spherical surface by $(\pi/4 \leq \theta \leq 3\pi/4) \cap (-\pi/2 \leq \phi \leq \pi/2)$. This grid could be effective when the processor number is multiple of three.

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